

SESSION 23

Differential Calculus

Rules of Differentiation · Partial Derivatives · Gradients · Jacobian & Hessian Matrices

Derivative as Tangent Slope

Fundamental Rules of Differentiation

The Chain Rule

Partial Derivatives

The Gradient Vector

The Jacobian Matrix

The Hessian Matrix

Taylor Series Approximation

Session Notes — Complete Reference

Playlist: Maths for Machine Learning · Instructor: Nitish Singh (CampusX)



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01 • Derivative as Tangent Slope

The derivative $f'(x)$ measures the instantaneous rate of change of a function $f(x)$ at point x , geometrically representing the slope of the tangent line to the function at that point.

02 • Fundamental Rules of Differentiation

Core rules of differentiation:

- **Power Rule:** $\frac{d(x^n)}{dx} = n x^{n-1}$.
- **Product Rule:** $\frac{d(uv)}{dx} = u'v + uv'$.
- **Quotient Rule:** $\frac{d(u/v)}{dx} = (u'v - uv') / v^2$.

03 • The Chain Rule

Essential for backpropagation in neural networks. It computes the derivative of composite functions:

CHAIN RULE

$$dy/dx = (dy/du) \cdot (du/dx)$$

04 • Partial Derivatives

When a function depends on multiple variables, the partial derivative $\frac{\partial f}{\partial x_i}$ measures the rate of change with respect to x_i while holding all other variables constant.

05 • The Gradient Vector

The **gradient** of a scalar function $f(\mathbf{x})$ is a vector containing all first-order partial derivatives of a scalar function:

GRADIENT VECTOR

$$\nabla f(\mathbf{x}) = \left[\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_n} \right]^\top$$

06 • The Jacobian Matrix

For a vector-valued function $\mathbf{f}: \mathbb{R}^n \rightarrow \mathbb{R}^m$, the **Jacobian** J is a matrix of all first-order partial derivatives, representing the local linear approximation of the function.

07 · The Hessian Matrix

The **Hessian** H is a square matrix of second-order partial derivatives of a scalar function, describing the local curvature (concavity/convexity) of the function:

HESSIAN MATRIX ELEMENTS

$$H_{(i,j)} = \partial^2 f / (\partial x_i \cdot \partial x_j)$$

08 · Taylor Series Approximation

A representation of a function as an infinite sum of terms calculated from the values of its derivatives at a single point, enabling us to approximate complex functions locally.

09 · Quick Reference Card



Calculus Cheat Sheet

- **Gradient:** Vector of first derivatives, points to steepest ascent.
- **Jacobian:** Matrix of first derivatives for vector functions.
- **Hessian:** Matrix of second derivatives, determines local curvature.